

ENTRY 032 Deriving and Shipping the Route-Then-Entangle Fix — August 13, 2026 — Status: COMPLETE

Entry 032 — The Control Qubit's Routing Errors Don't Matter

A Kraus-Operator Derivation, Verified by an Error Sweep That Shouldn't Move the Answer -- and Doesn't

1. The Derivation

Entry 031 found that most SWAP legs on a real route move UNENTANGLED product-state qubits, not an already-entangled payload. This entry works out exactly what that means. Call the qubit that receives the initial H the CONTROL (it carries |+> before any routing) and the qubit that starts at |0> the TARGET. Tracking the four combinations of "control corrupted / clean" × "target corrupted / clean" through the final entangling gate via its Kraus operators: if the control gets forced to a maximally mixed state before the final gate, it is now a CLASSICAL random bit (diagonal in the computational basis) -- combining a classical random control with a clean target through a CX/ECR still produces a definite, correlated pair (00 or 11), so it costs nothing. If the TARGET gets corrupted instead, the final gate combines a genuine coherent superposition with a classically randomized partner, producing a 50/50 mixture of a correct and an incorrect Bell state -- landing exactly at the floor. The two roles are not interchangeable.

Closed form for a single-pair circuit: $\text{success} = \text{target_survive} \times (1 - \text{final_raw}/2) + (1 - \text{target_survive}) \times 0.5$, where target_survive is the product of survival across only the target's routing legs and final_raw is the entangling gate's own error. In the no-routing limit ($\text{target_survive} = 1$) this reduces exactly to Entry 025's single-gate formula, $0.5 + 0.5(1-p)$.

2. Verified: Zero Sensitivity to the Control's Error

Sweeping the control-side edge's depolarizing error from 1% to 60% in an Aer simulation of the exact `bell_near_57_61` route structure left the measured success probability completely unchanged at every point (99.286%, to three decimal places) -- while this entry's closed form (evaluated without depending on that edge at all) matched it to within 0.11 points throughout. This is about as clean a confirmation as this kind of derivation gets.

3. Scope and Implementation

Implemented in `exact_dwll_cost()` for single-pair (plain Bell) circuits only -- 4 of the 5 circuits in every reference set used this session. Multi-pair (GHZ-style) circuits keep the old floor-based model unchanged, since later entangling gates there act on qubits already entangled from earlier gates, and the derivation above assumed a fresh product-state input -- extending it correctly would need its own derivation, not attempted here.

4. Results

circuit (chip)	Aer ref	gap before	gap after
<code>bell_adjacent_77_78</code> (Kyiv)	96.07%	0.09	0.15
<code>bell_near_77_82</code> (Kyiv)	48.74%	10.81	11.65
<code>bell_mid_77_100</code> (Kyiv)	83.69%	3.39	2.31
<code>bell_far_0_126</code> (Kyiv)	69.90%	11.21	6.07
<code>ghz_local_77_78_79</code> (Kyiv)	88.53%	0.10	0.10
KYIV MEAN / WORST		5.13 / 11.21	4.06 / 11.65
<code>bell_adjacent_0_1</code> (Sherbrooke)	95.53%	0.00	0.00
<code>bell_near_57_61</code> (Sherbrooke)	95.08%	15.98	1.65
<code>bell_mid_0_58</code> (Sherbrooke)	49.40%	17.57	20.19
<code>bell_far_0_126</code> (Sherbrooke)	49.51%	14.90	20.39
<code>ghz_local_0_1_14</code> (Sherbrooke)	91.48%	1.05	1.05
SHERBROOKE MEAN / WORST		9.90 / 17.57	8.66 / 20.39

Entry 032 — Conclusion

Mean gap improved on both chips (Kyiv 5.13 → 4.06, Sherbrooke 9.90 → 8.66), and bell_near_57_61 -- the specific circuit this whole investigation started from -- dropped from a 15.98-point gap to 1.65, close to shot noise. That is a genuine, derived, twice-verified fix, and it ships as of this entry. It is not a clean sweep, though: bell_mid and bell_far got WORSE on Sherbrooke (17.57 → 20.19 and 14.90 → 20.39) and bell_near got marginally worse on Kyiv (10.81 → 11.65). Removing gate-error charges from the control's path made the model more optimistic everywhere that charge used to apply -- correctly so, per the derivation -- but on the very longest routes, the real measured success is low enough that removing ANY pessimistic term moves the prediction further from reality. That is not evidence the fix is wrong; the control-sweep test proves it isn't. It is evidence that bell_mid and bell_far have a separate, still-unidentified source of loss big enough that it was previously masked by an unrelated bug's over-charging.

5. Next Step — Entry 033

Isolate bell_far_0_126 and bell_mid_0_58/bell_mid_77_100 the same way this entry isolated bell_near_57_61: ablate gate error, decoherence, and readout separately against Aer for these SPECIFIC long routes now that the route-then-entangle and control/target asymmetry are both accounted for, to find what's actually left unmodeled. Candidates worth checking first: whether the TARGET's own long chain has a similar asymmetry still unaccounted for once it spans many hops, or whether something entirely new (crosstalk between simultaneously-active spectator qubits on a 26-hop route, for instance) only becomes visible at this route length.

Research Log — Index Addendum (Entry 032)

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Entry 032	Route-Then-Entangle Fix Shipped: Control-Side Gate Error Doesn't Matter	Aug 13, 2026	Complete

exact_dwelling_routing.py's exact_dwelling_cost() now splits single-pair gate-error handling into a target-survival closed form, verified via a control-edge sensitivity sweep. GHZ/multi-pair circuits unchanged. bell_mid/bell_far remain open (Entry 033).